Project 2 — Loan Default Prediction: Results & Analysis

Logistic Regression on Kaggle "Give Me Some Credit" Dataset

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Executive Summary

A Logistic Regression model was trained to predict whether a borrower will default on a loan within 2 years. The model achieved an **AUC-ROC of 0.8334**, meaning it correctly ranks a real defaulter above a non-defaulter 83% of the time — a strong result for a linear model on messy consumer credit data. It caught **74% of all actual defaults** (Recall = 0.74) in the held-out test set, missing 1,939 real defaulters out of 7,461.

Dataset Summary

Property	Value
Source	Kaggle — Give Me Some Credit
Raw rows	150,000 borrowers
After cleaning (NaN removal, outlier capping)	120,269 rows
Natural default rate	6.68% (heavily imbalanced)
After upsampling minority class	149,216 rows at 25% default rate
Train set	119,372 rows (80%)
Test set	29,844 rows (20%)

Why the Dataset Was Rebalanced

The natural 6.68% default rate means a model that predicts "No Default" for every single borrower would still achieve 93.3% accuracy — while catching zero real defaulters. That model would be worthless for a bank. To force the model to actually learn default patterns, the minority class (defaults) was upsampled to 25% of the training data using random oversampling with replacement. The test set was kept at its natural ratio — oversampling test data would give falsely optimistic results.

Features Used

Feature	Description	Domain Logic
DTI (Debt Ratio)	Monthly debt payments / monthly income	High DTI = stretched borrower, less room for shocks
Age	Borrower age in years	Older borrowers historically default less (more stable income)
Late_30_59	Times 30–59 days past due	Recent stress signal — most predictive when recent
Late_60_89	Times 60–89 days past due	Serious delinquency — escalating pattern
Late_90	Times 90+ days past due	Most severe — strongest individual default predictor
Revolving_util	Credit card utilisation rate	>90% utilisation = borrower is maxed out
Monthly_income	Monthly income in USD	Low income = lower buffer against income shocks
Open_credit_lines	Number of open credit accounts	Excessive credit-seeking behaviour is a risk signal

Outlier capping at the 99th percentile was applied to Revolving_util, DTI, and Monthly_income. Without this, a single borrower with DTI = 3,000 or utilisation = 40,000 would dominate the linear model's coefficient estimates.

Model: Logistic Regression

Logistic Regression is the global industry standard for credit scoring — it underlies FICO scores, bank scorecards, and Basel III risk models. It outputs a **probability of default** between 0 and 1 rather than a hard label.

The decision function is:

$$P(\text{default}) = \frac{1}{1 + e^{-(w_1 \cdot \text{DTI} + w_2 \cdot \text{Age} + w_3 \cdot \text{Late_30_59} + \ldots + b)}}$$

A predicted probability above 0.50 is classified as Default. The parameter `class_weight="balanced"` adds an additional penalty for missing real defaults during training, pushing the model to be more sensitive to the minority class even after upsampling.

Results

Core Metrics

Metric	Value	Interpretation
AUC-ROC	0.8334	Ranks defaulters above non-defaulters 83% of the time
Avg Precision	0.6572	2.6× above the random baseline of 0.25
Overall Accuracy	76%	Misleading — only 1% above the naive "always say No Default" baseline

Classification Report

Class	Precision	Recall	F1-Score	Support
No Default	0.90	0.77	0.83	22,383
Default	0.52	0.74	0.61	7,461
Weighted Avg	0.80	0.76	0.77	29,844

Confusion Matrix

	Predicted: No Default	Predicted: Default
Actual: No Default	TN = 17,244 ✅	FP = 5,139 ⚠️
Actual: Default	FN = 1,939 ❌	TP = 5,522 ✅

Interpreting Each Number

AUC-ROC = 0.8334

AUC measures ranking quality — if you take one random defaulter and one random non-defaulter from the test set, the model assigns a higher probability of default to the real defaulter 83.34% of the time. A random classifier scores 0.50. A perfect model scores 1.0. In consumer credit risk, anything above 0.75 is considered production-grade for a baseline model.

Precision vs Recall Tradeoff on Defaults

Recall (0.74) is the metric a bank's credit risk team cares about most. It answers: of all the borrowers who actually defaulted, what fraction did the model flag? A Recall of 0.74 means **the model caught 5,522 out of 7,461 real defaulters** — and missed 1,939.

Precision (0.52) answers the opposite question: of all the borrowers flagged as likely defaulters, how many actually defaulted? A Precision of 0.52 means **roughly half of the flagged borrowers were false alarms** (5,139 good customers incorrectly rejected). This is the cost of high recall — to catch more real defaults, the model also flags more innocent borrowers.

This tradeoff is tunable by adjusting the classification threshold. Lowering the threshold from 0.50 to 0.30 would catch more real defaults (higher Recall) but generate even more false alarms (lower Precision).

False Negatives = 1,939 — The Expensive Number

A False Negative means the model said "this borrower is safe" but they defaulted. The bank approves the loan, the borrower defaults, and the bank loses the principal. This is the most costly error in credit risk and is the primary target for improvement.

False Positives = 5,139 — The Revenue Cost

A False Positive means the model flagged a good borrower as a default risk. The bank rejects their application. The borrower is creditworthy but denied a loan. This costs the bank interest income and damages the customer relationship, but does not result in principal loss.

Why Accuracy = 76% Is Misleading

A naive model that predicts "No Default" for every single borrower would achieve 75% accuracy on this balanced test set (since 75% of the test set is non-defaulters). The logistic regression model achieves 76% — barely above that naive baseline. If accuracy were the only metric reported, this model would look barely better than guessing. The correct story is told by AUC (0.83) and Default Recall (0.74), which confirm the model has genuine predictive power on the class that matters.

Feature Coefficients — What Drives Default Risk

The sign and magnitude of each coefficient reveals the model's learned credit risk logic:

Feature	Effect on Default Probability	Business Interpretation
Late_90	Strongly increases ↑	90+ day delinquency is the single strongest default predictor
Late_60_89	Increases ↑	Serious past delinquency compounds risk
Late_30_59	Increases ↑	Even moderate past-due history raises risk
Revolving_util	Increases ↑	Maxed-out credit cards = cash-flow distress
DTI	Increases ↑	Debt-heavy borrowers have no buffer for income shocks
Age	Decreases ↓	Older borrowers have more stable income and credit history
Monthly_income	Decreases ↓	Higher income = more buffer, lower default probability
Open_credit_lines	Near zero	Weak signal — number of lines alone doesn't predict default

The late payment features (Late_90, Late_60_89, Late_30_59) dominating the coefficient list is consistent with decades of credit scoring research: **the best predictor of future delinquency is past delinquency.**

Charts Produced

confusion_matrix.png

The 2×2 heatmap shows the raw count of each prediction outcome. The bottom-left cell (FN = 1,939) is the most operationally important number — it represents real defaulters the model failed to catch.

roc_curve.png

The ROC curve plots True Positive Rate vs False Positive Rate at every possible classification threshold. The blue curve bowing toward the top-left corner — far above the red diagonal (random classifier) — confirms the model's discriminative ability. The area under this curve is 0.8334.

precision_recall_curve.png

The Precision-Recall curve plots Precision vs Recall as the threshold changes. This is the more honest metric for imbalanced classification because it focuses entirely on the positive class (defaults). The red dashed baseline represents a random classifier at the test set's natural default rate. The model's curve sits well above this baseline, confirming real skill on the minority class.

feature_coefficients.png

Horizontal bar chart of standardised coefficients. Red bars indicate features that increase default probability; blue bars indicate features that decrease it. The late payment features dominate, with age and income as protective factors.

Limitations

1. **Linear decision boundary** — Logistic Regression cannot capture non-linear interactions. For example, the combination of high DTI *and* high utilisation *and* recent late payments may be far more dangerous than each factor alone. A tree-based model (XGBoost) would capture these interaction effects.
2. **No temporal structure** — The model treats every borrower identically regardless of economic cycle. Loans originated during a recession have higher default rates than identical-feature loans during expansion. This vintage effect is not captured.
3. **Threshold is fixed at 0.50** — The optimal threshold depends on the bank's specific cost structure (cost of FN vs cost of FP). A bank that loses ₹50L on each missed default but only loses ₹1L on each false alarm should lower the threshold significantly.
4. **Upsampling on training data only** — Correct methodology, but the model's real-world performance should be evaluated on the natural 6.68% default rate, not the balanced 25% rate.

Next Steps

Step	Expected Improvement
XGBoost on same dataset	AUC from 0.83 → 0.87–0.89
Threshold optimisation at bank's cost ratio	Reduce FN at controlled FP increase
Add loan amount, interest rate features	Contextualise risk by loan size

Step	Expected Improvement
Walk-forward validation by origination year	Capture vintage effects
SHAP values for explainability	Understand individual predictions

Project 2 of the NSE/Credit ML Series. Logistic Regression baseline established. Moving to XGBoost next.